

Signals and Systems

Complete Lecture Notes

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I. Signals

1. Signal Classification

- Time: Continuous vs. Discrete
- Continuous: Analog (e.g., continuous sampling)
- Discrete: Digital (quantized sequence)

2. Signal Decomposition

- ① Convergence Property:
- Continuous-time non-periodic (Fourier Transform):

$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) e^{j\omega t} d\omega$$

Example: $t \leftrightarrow \delta(t)$, Laplace Transform kernel e^{-st} , Fourier Transform kernel $e^{-j\omega t}$

- Continuous-time periodic (Fourier Series):

$$f(t) = \sum_{n=-\infty}^{\infty} F_n e^{jn\omega_0 t}$$

Example: Fourier coefficients, Laplace Transform, Fourier Transform

- Discrete-time non-periodic (DTFT):

$$f[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} F(e^{j\omega}) e^{j\omega n} d\omega$$

Example: $n \leftrightarrow \delta[n]$, z-Transform kernel z^{-n} , DTFT kernel $e^{-j\omega n}$

- Discrete-time periodic (DFS):

$$f[n] = \sum_{k=0}^{N-1} \tilde{X}[k] e^{j\frac{2\pi}{N}kn}$$

Example: DFS coefficients, z-Transform, DFT

- ② Orthogonality:
- Inner product of signals: $\langle \phi_n(t), f(t) \rangle = \int_{-\infty}^{\infty} f(t) \phi_n^*(t) dt$
- Discrete signals inner product: $\langle f[n], g[n] \rangle = \sum_{n=-\infty}^{\infty} f[n] g^*[n]$
- Orthogonality relation: $\sum_n \phi_m[n] \phi_n^*[n] = \lambda_n \delta_{mn}$

Energy Signals:

- Cross-correlation: $R_{21}(t) = \langle f_2(\tau), f_1(\tau - t) \rangle = R_{12}^*(-t)$
- Total energy: $E = R_{11}(0) = \int_{-\infty}^{\infty} |f(t)|^2 dt$

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Power Signals:

- Average power: $P = R_{11}(0) = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} |f(t)|^2 dt$
- Autocorrelation: $R_{ff}(\tau) = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} f(t) f^*(t - \tau) dt$

II. Systems

1. System Properties

- Superposition (Linearity): $\mathcal{H}\{ax_1(t) + bx_2(t)\} = a\mathcal{H}\{x_1(t)\} + b\mathcal{H}\{x_2(t)\}$
- Time-invariance: If $x(t) \rightarrow y(t)$, then $x(t - t_0) \rightarrow y(t - t_0)$
- Causality: Output depends only on present and past inputs.
- Stability: BIBO (Bounded-Input Bounded-Output) stable.

2. Impulse Response of Combined LTI Systems

- Parallel Connection: $h(t) = h_1(t) + h_2(t)$
- Cascade Connection: $h(t) = h_1(t) * h_2(t)$

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III. Time-Domain Analysis of LTI Systems

1. Differential & Difference Equation Formulations

- Continuous LTI systems:

$$\sum_{k=0}^N a_k \frac{d^k y(t)}{dt^k} = \sum_{m=0}^M b_m \frac{d^m x(t)}{dt^m}$$

- Discrete LTI systems:

$$\sum_{k=0}^N a_k y[n - k] = \sum_{m=0}^M b_m x[n - m]$$

2. Convolution Integrals & Sums

- Convolution Integral:

$$y(t) = x(t) * h(t) = \int_{-\infty}^{\infty} x(\tau) h(t - \tau) d\tau$$

- Convolution Sum:

$$y[n] = x[n] * h[n] = \sum_{k=-\infty}^{\infty} x[k] h[n - k]$$

3. Impulse Response matching method

Used to find initial conditions at $t = 0^+$ from $t = 0^-$ when an impulse input $\delta(t)$ is applied.

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IV. Laplace Transform and s-Domain Analysis

1. Definition of Bilateral Laplace Transform

$$X(s) = \int_{-\infty}^{\infty} x(t)e^{-st}dt, \quad s = \sigma + j\omega$$

2. Region of Convergence (ROC) properties

- Determined by $\text{Re}(s) = \sigma$.
- For right-sided signals, ROC is $\text{Re}(s) > \sigma_{\max}$.
- For left-sided signals, ROC is $\text{Re}(s) < \sigma_{\min}$.
- For two-sided signals, ROC is a strip $\sigma_1 < \text{Re}(s) < \sigma_2$.

3. Partial Fraction Expansion (PFE)

Real poles: $H(s) = \sum_{i=1}^N \frac{r_i}{s-p_i} \leftrightarrow h(t) = \sum_{i=1}^N r_i e^{p_i t} u(t)$

Conjugate complex poles: $H(s) = \frac{As+B}{(s+\alpha)^2+\beta^2} \leftrightarrow h(t) = Ce^{-\alpha t} \cos(\beta t + \theta)u(t)$

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Laplace Transform Properties

Property	Time Domain $x(t)$	s-Domain $X(s)$	ROC
Linearity	$ax_1(t) + bx_2(t)$	$aX_1(s) + bX_2(s)$	Contains $R_1 \cap R_2$
Time Shifting	$x(t - t_0)$	$e^{-st_0} X(s)$	Same as R
s-Domain Shifting	$e^{s_0 t} x(t)$	$X(s - s_0)$	Shifted by $\text{Re}(s_0)$
Scaling	$x(at)$	$\frac{1}{ a } X\left(\frac{s}{a}\right)$	Scaled by a
Differentiation	$\frac{dx(t)}{dt}$	$sX(s) - x(0^-)$	Contains R
Integration	$\int_{-\infty}^t x(\tau) d\tau$	$\frac{X(s)}{s}$	Contains $R \cap \{\text{Re}(s) > 0\}$
Convolution	$x_1(t) * x_2(t)$	$X_1(s)X_2(s)$	Contains $R_1 \cap R_2$

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V. Fourier Analysis of Continuous-Time Signals

1. Energy and Power Signal Spectrum

- Energy spectral density: $S_e(j\omega) = |X(j\omega)|^2$, with total energy:

$$E_y = \frac{1}{2\pi} \int_{-\infty}^{\infty} |X(j\omega)|^2 d\omega$$

- Power spectral density: $S_p(j\omega) = \lim_{T \rightarrow \infty} \frac{|X_T(j\omega)|^2}{T}$, with average power:

$$P_y = \frac{1}{2\pi} \int_{-\infty}^{\infty} S_p(j\omega) d\omega$$

2. LTI System Transfer Function

- Reciprocity: $H(j\omega)H(-j\omega) = 1$
- Causality: ROC of rational $H(s)$ lies in the right-half plane $\text{Re}(s) > \sigma_0$.
- Stability: BIBO stable if ROC contains the $j\omega$ axis ($\text{Re}(s) = 0$).

Fourier Transform of Periodic Signals

If $f(t)$ is periodic with period T , then we can write it using the Fourier series representation:

$$f(t) = \sum_{n=-\infty}^{\infty} F_n e^{jn\omega_0 t}, \quad \omega_0 = \frac{2\pi}{T}$$

Evaluating its Fourier Transform yields:

$$F(\omega) = 2\pi \sum_{n=-\infty}^{\infty} F_n \delta(\omega - n\omega_0)$$

where the coefficients F_n are given by:

$$F_n = \frac{1}{T} \int_{-T/2}^{T/2} f(t) e^{-jn\omega_0 t} dt$$

For periodic power signals, the autocorrelation spectral density satisfies:

$$R_f(\tau) = \sum_{n=-\infty}^{\infty} |F_n|^2 e^{jn\omega_0 \tau} \leftrightarrow S_f(\omega) = 2\pi \sum_{n=-\infty}^{\infty} |F_n|^2 \delta(\omega - n\omega_0)$$

VI. Discrete-Time LTI System Response

1. Time Domain Representation

$$x[n] = \sum_{m=-\infty}^{\infty} x[m] \delta[n - m]$$

2. Response Decomposition

Total response is decomposed into the homogeneous solution (impulse response) and the particular solution (forced response):

$$y[n] = y_h[n] + y_p[n]$$

VII. The Hilbert Transform

1. Continuous-Time Definition

$$\hat{x}(t) = \mathcal{H}\{x(t)\} = x(t) * \frac{1}{\pi t} = \frac{1}{\pi} \int_{-\infty}^{\infty} \frac{x(\tau)}{t - \tau} d\tau$$

In the frequency domain, the transfer function is:

$$H(j\omega) = -j \operatorname{sgn}(\omega)$$

2. Properties

An analytic signal is defined as: $z(t) = x(t) + j\hat{x}(t)$. Its spectrum is double-sided, containing only positive frequencies:

$$Z(j\omega) = \begin{cases} 2X(j\omega), & \omega > 0 \\ X(j\omega), & \omega = 0 \\ 0, & \omega < 0 \end{cases}$$

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VIII. The z-Transform

1. Bilateral z-Transform

$$X(z) = \sum_{n=-\infty}^{\infty} x[n]z^{-n}$$

ROC: the set of values z for which $\sum_{n=-\infty}^{\infty} |x[n]z^{-n}| < \infty$ converges.

- Right-sided signal: ROC is $|z| > r_0$
- Left-sided signal: ROC is $|z| < r_0$
- Two-sided signal: ROC is an annular region $r_i < |z| < r_o$

2. Rational Transfer Functions & Pole locations

For causal real-pole and complex-pole systems:

$$H(z) = \sum_{i=1}^P \frac{A_i}{1 - p_i z^{-1}} \leftrightarrow h[n] = \sum_{i=1}^P A_i p_i^n u[n]$$

3. System Stability

- Stable: ROC of $H(z)$ contains the unit circle $|z| = 1$.
- Causal: ROC is the exterior of a circle $|z| > r_{\max}$.

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z-Transform Properties

1. Line multiplication: $n^m x[n] \leftrightarrow \left(-z \frac{d}{dz}\right)^m X(z)$
2. Time Shifting: $x[n - n_0] \leftrightarrow z^{-n_0} X(z)$
3. Frequency Scaling (Rotation): $a^n x[n] \leftrightarrow X(a^{-1}z)$
4. Time Reversal: $x[-n] \leftrightarrow X(z^{-1})$
5. Conjugation: $x^*[n] \leftrightarrow X^*(z^*)$
6. Convolution: $x_1[n] * x_2[n] \leftrightarrow X_1(z)X_2(z)$
7. Initial Value Theorem: If $x[n]$ is causal, $x[0] = \lim_{z \rightarrow \infty} X(z)$
8. Final Value Theorem: If $x[n]$ is causal and stable, $x[\infty] = \lim_{z \rightarrow 1} (z - 1)X(z)$

Single-sided z-Transform properties:

- Time Shifting (Advance): $x[n + m]u[n] \leftrightarrow z^m \left[X(z) - \sum_{k=0}^{m-1} x[k]z^{-k} \right]$
 - Time Shifting (Delay): $x[n - m]u[n] \leftrightarrow z^{-m} \left[X(z) + \sum_{k=-m}^{-1} x[k]z^{-k} \right]$
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IX. Discrete-Time Fourier Transform (DTFT)

1. Definition

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]e^{-jn\omega}$$

2. Relation to z-Transform

If the ROC of $X(z)$ contains the unit circle ($|z| = 1$), the DTFT is equivalent to evaluating the z-transform on the unit circle:

$$X(e^{j\omega}) = X(z) \Big|_{z=e^{j\omega}}$$

3. LTI System output DTFT

$$Y(e^{j\omega}) = X(e^{j\omega})H(e^{j\omega})$$

DTFT of Periodic Sequences

Given periodic sequence $x[n]$ with period N , its Discrete Fourier Series (DFS) representation is:

$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k]e^{j\frac{2\pi}{N}kn}$$

Its DTFT is represented as a train of impulse deltas in the frequency domain:

$$X(e^{j\omega}) = \frac{2\pi}{N} \sum_{k=-\infty}^{\infty} X[k]\delta\left(\omega - \frac{2\pi k}{N}\right)$$

where the DFS/DFT coefficients $X[k]$ are:

$$X[k] = \sum_{n=0}^{N-1} x[n]e^{-j\frac{2\pi}{N}kn}$$

For a finite length sequence of length N_0 , if we extend it periodically to $x_N[n]$ with period $N \geq N_0$, the DTFT samples are directly related to the DFT coefficients:

$$X_N[k] = X(e^{j\omega}) \Big|_{\omega=\frac{2\pi}{N}k}$$

X. Sampling in Time and Frequency

1. Time-Domain Sampling of Continuous Signals

Continuous signal $f(t)$ sampled with interval T gives $x[n] = f(nT)$. In the frequency domain:

$$X(e^{j\omega}) = \frac{1}{T} \sum_{m=-\infty}^{\infty} F_a \left(j \left(\frac{\omega}{T} - m \frac{2\pi}{T} \right) \right)$$

If the continuous spectrum is bandlimited to ω_m and the sampling frequency satisfies the Nyquist rate $\omega_s = \frac{2\pi}{T} > 2\omega_m$, reconstruction is possible using ideal interpolation:

$$f(t) = \sum_{n=-\infty}^{\infty} f(nT) \text{sinc}\left(\frac{t - nT}{T}\right)$$

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XI. The Fast Fourier Transform (FFT)

1. Decimation-in-Time Base-2 FFT (DIT-FFT)

Splitting the DFT sum into even and odd indices:

$$X[k] = \sum_{r=0}^{N/2-1} x[2r] W_{N/2}^{kr} + W_N^k \sum_{r=0}^{N/2-1} x[2r+1] W_{N/2}^{kr}$$

$$X[k + N/2] = \sum_{r=0}^{N/2-1} x[2r] W_{N/2}^{kr} - W_N^k \sum_{r=0}^{N/2-1} x[2r+1] W_{N/2}^{kr}$$

2. Decimation-in-Frequency Base-2 FFT (DIF-FFT)

Splitting the output sequence into even and odd bins:

$$X[2r] = \sum_{n=0}^{N/2-1} (x[n] + x[n + N/2]) W_{N/2}^{nr}$$

$$X[2r+1] = \sum_{n=0}^{N/2-1} (x[n] - x[n + N/2]) W_N^n W_{N/2}^{nr}$$

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XII. Decimation and Interpolation

1. Decimation (Downsampling by M)

Anti-aliasing low-pass filtering followed by downsampling:

$$y[n] = v[nM] \leftrightarrow Y(e^{j\omega}) = \frac{1}{M} \sum_{i=0}^{M-1} V\left(e^{j\frac{\omega-2\pi i}{M}}\right)$$

2. Interpolation (Upsampling by L)

Upsampling by inserting $L - 1$ zeros followed by interpolation low-pass filtering:

$$w[n] = \begin{cases} x[n/L], & n = 0, \pm L, \pm 2L, \dots \\ 0, & \text{otherwise} \end{cases} \leftrightarrow W(e^{j\omega}) = X(e^{j\omega L})$$

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XIII. Filter Structures and Lossless Network Synthesis

1. Lossless Double-Terminated Networks

For a continuous network with source resistance R_s and load resistance R_L , the rated power of the source is $P_{\max} = \frac{|E_s|^2}{4R_s}$. The power transmission coefficient is:

$$|H(j\Omega)|^2 = \frac{P_L}{P_{\max}} = 4 \frac{R_s}{R_L} \left| \frac{V_o(j\Omega)}{E_s} \right|^2$$

The reflection coefficient function satisfies:

$$|\rho(j\Omega)|^2 = 1 - |H(j\Omega)|^2$$

2. Digital Filter Realization (Difference Equations)

Direct realization of IIR systems:

$$H(z) = \frac{\sum_{k=0}^M b_k z^{-k}}{1 + \sum_{k=1}^N a_k z^{-k}} \leftrightarrow y[n] = \sum_{k=0}^M b_k x[n-k] - \sum_{k=1}^N a_k y[n-k]$$

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Butterworth Filter Design

1. Magnitude squared frequency response

$$|H_a(j\Omega)|^2 = \frac{1}{1 + \left(\frac{\Omega}{\Omega_c}\right)^{2N}}$$

2. Pole distributions

$$s_k = \Omega_c e^{j\left[\frac{\pi}{2} + \frac{2k-1}{2N}\pi\right]}, \quad k = 1, 2, \dots, 2N$$

3. Stable lowpass transfer function

Selecting only LHP (left-half plane) poles to ensure stability:

$$H_a(s) = \frac{\Omega_c^N}{\prod_{k \in \text{LHP}} (s - s_k)} = \frac{\Omega_c^N}{B_N(s')}$$

where $s' = s/\Omega_c$ is the normalized Laplace variable, and $B_N(s')$ is the normalized Butterworth polynomial.

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XIV. Infinite Impulse Response (IIR) Digital Filter Design

1. Impulse Invariance Method

The continuous impulse response is sampled to form the discrete system: $h[n] = T_d h_a(nT_d)$.

$$\frac{A}{s - s_0} \Rightarrow \frac{AT_d}{1 - e^{s_0 T_d} z^{-1}}$$

Conclusion: Suffers from aliasing at high frequencies.

2. Bilinear Transform Method

A mapping that maps the continuous s-plane to the discrete z-plane without aliasing:

$$s = \frac{2}{T_d} \left(\frac{1 - z^{-1}}{1 + z^{-1}} \right)$$

Frequency warping relation:

$$\Omega = \frac{2}{T_d} \tan \left(\frac{\omega}{2} \right)$$

Conclusion: Free of aliasing, but exhibits non-linear frequency warping distortion.

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XV. Finite Word Length Effects in Digital Filters

1. Quantization Error Models

- Rounding (uniform on $[-q/2, q/2]$): Mean $\mu_e = 0$, Variance $\sigma_e^2 = \frac{q^2}{12}$
- Truncation (uniform on $[-q, 0]$): Mean $\mu_e = -q/2$, Variance $\sigma_e^2 = \frac{q^2}{12}$

2. Output Noise Power

For quantization noise passing through an LTI system with impulse response $h[n]$:

$$\sigma_{yo}^2 = \sigma_e^2 \sum_{n=-\infty}^{\infty} |h[n]|^2 = \sigma_e^2 \frac{1}{2\pi} \int_{-\pi}^{\pi} |H(e^{j\omega})|^2 d\omega$$

3. Scaling to Avoid Overflow

To ensure node variables $w[n]$ do not exceed 1, we apply a scaling factor β to the input signal:

$$\beta \leq \frac{1}{\sum_{n=-\infty}^{\infty} |f[n]|} \quad (L_1 \text{ bound})$$

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XVI. Linear Phase FIR Filter Design

1. Symmetry and Anti-symmetry Conditions

Symmetric (Type 1 & 2): $h[n] = h[N - 1 - n]$

Anti-symmetric (Type 3 & 4): $h[n] = -h[N - 1 - n]$

2. Frequency Response $H(e^{j\omega})$ Types

- Type 1 (Symmetric, N odd): $H(e^{j\omega}) = e^{-j\omega(N-1)/2} \sum_{n=0}^{(N-1)/2} a[n] \cos(\omega n)$
- Type 2 (Symmetric, N even): $H(e^{j\omega}) = e^{-j\omega(N-1)/2} \sum_{n=1}^{N/2} b[n] \cos \left(\omega \left(n - \frac{1}{2} \right) \right)$

3. Frequency Transformation of Lowpass Digital Filters

Lowpass mapping function for the unit circle:

$$Z^{-1} = G(z^{-1}) = \pm \frac{z^{-1} - \alpha}{1 - \alpha z^{-1}}$$

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XVII. State-Space Representation of Systems

1. Continuous-Time State-Space Equations

$$\dot{\boldsymbol{x}}(t) = \boldsymbol{A}\boldsymbol{x}(t) + \boldsymbol{B}\boldsymbol{u}(t)$$

$$\boldsymbol{y}(t) = \boldsymbol{C}\boldsymbol{x}(t) + \boldsymbol{D}\boldsymbol{u}(t)$$

- Time domain solution:

$$\boldsymbol{x}(t) = e^{\boldsymbol{A}t}\boldsymbol{x}(0) + \int_0^t e^{\boldsymbol{A}(t-\tau)}\boldsymbol{B}\boldsymbol{u}(\tau)d\tau$$

- s-domain solution:

$$\boldsymbol{X}(s) = (s\boldsymbol{I} - \boldsymbol{A})^{-1}\boldsymbol{x}(0) + (s\boldsymbol{I} - \boldsymbol{A})^{-1}\boldsymbol{B}\boldsymbol{U}(s)$$

$$\boldsymbol{Y}(s) = \boldsymbol{C}(s\boldsymbol{I} - \boldsymbol{A})^{-1}\boldsymbol{x}(0) + [\boldsymbol{C}(s\boldsymbol{I} - \boldsymbol{A})^{-1}\boldsymbol{B} + \boldsymbol{D}]\boldsymbol{U}(s)$$

2. Discrete-Time State-Space Equations

$$\boldsymbol{x}[n+1] = \boldsymbol{A}\boldsymbol{x}[n] + \boldsymbol{B}\boldsymbol{u}[n]$$

$$\boldsymbol{y}[n] = \boldsymbol{C}\boldsymbol{x}[n] + \boldsymbol{D}\boldsymbol{u}[n]$$

- Time domain solution:

$$\boldsymbol{x}[n] = \boldsymbol{A}^n\boldsymbol{x}[0] + \sum_{k=0}^{n-1} \boldsymbol{A}^{n-1-k}\boldsymbol{B}\boldsymbol{u}[k]$$

- z-domain solution:

$$\boldsymbol{X}(z) = z(\boldsymbol{I} - \boldsymbol{A})^{-1}\boldsymbol{x}[0] + (\boldsymbol{I} - \boldsymbol{A})^{-1}\boldsymbol{B}\boldsymbol{U}(z)$$

$$\boldsymbol{Y}(z) = z\boldsymbol{C}(\boldsymbol{I} - \boldsymbol{A})^{-1}\boldsymbol{x}[0] + [\boldsymbol{C}(\boldsymbol{I} - \boldsymbol{A})^{-1}\boldsymbol{B} + \boldsymbol{D}]\boldsymbol{U}(z)$$

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State-Space Linear Coordinate Transformation

1. Coordinate Mapping

Let $\boldsymbol{x}' = \boldsymbol{P}\boldsymbol{x}$, where $\boldsymbol{P}$ is a non-singular transformation matrix. The transformed matrices are:

$$\boldsymbol{A}' = \boldsymbol{P}\boldsymbol{A}\boldsymbol{P}^{-1}, \quad \boldsymbol{B}' = \boldsymbol{P}\boldsymbol{B}, \quad \boldsymbol{C}' = \boldsymbol{C}\boldsymbol{P}^{-1}$$

2. Controllability and Observability Criteria

- An N -th order system is controllable if the controllability matrix $\boldsymbol{Q}_c$ has rank N :

$$\boldsymbol{Q}_c = [\boldsymbol{B}, \boldsymbol{A}\boldsymbol{B}, \boldsymbol{A}^2\boldsymbol{B}, \dots, \boldsymbol{A}^{N-1}\boldsymbol{B}]$$

- An N -th order system is observable if the observability matrix $\boldsymbol{Q}_o$ has rank N :

$$\boldsymbol{Q}_o = \begin{bmatrix} \boldsymbol{C} \\ \boldsymbol{C} \\ \boldsymbol{A} \\ \boldsymbol{C} \\ \boldsymbol{A}^2 \\ \vdots \\ \boldsymbol{C} \\ \boldsymbol{A}^{N-1} \end{bmatrix}$$
